NUQA

A Cleaner City, Through Intelligence

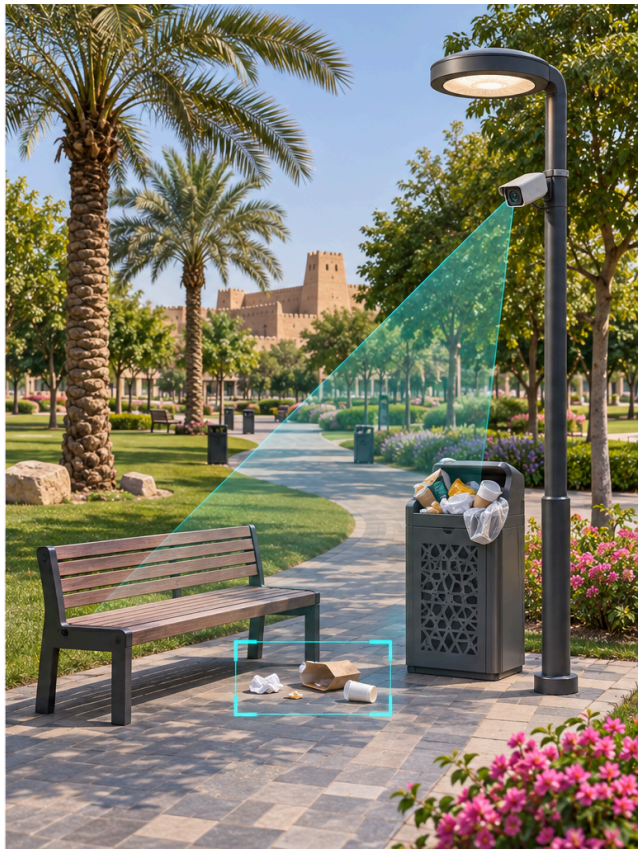
Connect observation, response and verified action.

SMART CAMERAS + CONNECTED PEOPLE



An ordinary afternoon. A problem that keeps growing.

Illustrative scenario: one overflowing park bin affects visitors, the environment and the cleaning team.



01 A family arrives

The nearest bin is full. Waste has collected beside the seating area.

02 The impact spreads

Visitors avoid the area; loose litter can reach planted ground or drains.

03 The team lacks a clear signal

Without a timely, located alert, staff must find and prioritize the cleanup.

The unmet need: detect it, locate it and give someone clear responsibility.

The scale behind the scene: global plastic-waste data

OECD data for 2019, published in 2022. Percentages refer to global plastic waste by treatment route.

22%

Mismanaged globally

9%

Recycled globally

Why it matters

Uncontrolled waste can reach soil and waterways, harm wildlife and reduce the quality of shared places.

Global plastic-waste treatment mix



Source: OECD, Global Plastics Outlook (2022), data year 2019. doi.org/10.1787/de747aef-en

The problem NUQA is designed to address

Litter can harm the environment. NUQA connects detection to a coordinated response.

01

Environmental harm

Litter can contaminate soil and water, harm wildlife and degrade shared green spaces.

02

Littering lacks context

A sighting alone may not reliably establish when, where or from which vehicle waste was thrown.

03

Response is fragmented

Disconnected reports, assignments and updates make ownership and closure harder to follow.

What NUQA targets — and how we will measure it

The project focuses on detection and operational response. Pilot results will determine its effect.

CHALLENGE	NUQA ACTION	PILOT MEASURE
Scattered litter	AI flags visible waste around paths, bins and seating.	Persistent litter per site
Overflowing bins	Review overflow alerts and assign collection tasks.	Time a bin stays overflowing
Vehicle littering	Link the event and vehicle evidence for staff review.	Valid events and attribution errors
Delayed follow-up	Unify reports, ownership and verified cleanup.	Response and closure time

Intended benefit: less persistent litter, cleaner shared places and a clearer response process.

The solution: one connected cleanliness platform

NUQA brings smart-camera alerts, citizen reports and field work into a shared case record.

01

Observe

Camera signals + citizen reports
Location, time and event context.

02

Coordinate

Staff review + task assignment
A clear owner and next action.

03

Resolve

Field response + verification
Citizens receive status updates.

The intended result: earlier awareness, clearer responsibility and traceable resolution.

Artificial intelligence analyzes the scene

Proposed edge-AI processing turns camera frames into a structured event for review.

01

Video frames

Capture the approved road or public-space zone.

02

Detect and track

Locate waste, bins and relevant object movement.

03

Classify the event

Flag overflow, scattered litter or a potential throwing event.

04

Create an alert

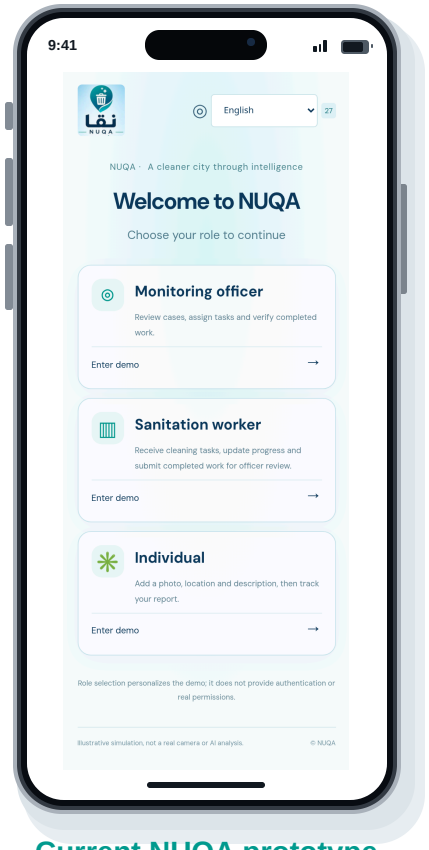
Attach event type, time, location and selected evidence.

Roads: link a potential throwing event to the vehicle; read a visible plate when permitted.

Parks: analyze waste around bins, benches and paths, then route the event for staff review.

Three roles. A focused experience for each.

The working website starts with role selection and offers 27 interface languages.



Current NUQA prototype

01 Individual

Report a location and issue; follow its status.

02 Sanitation worker

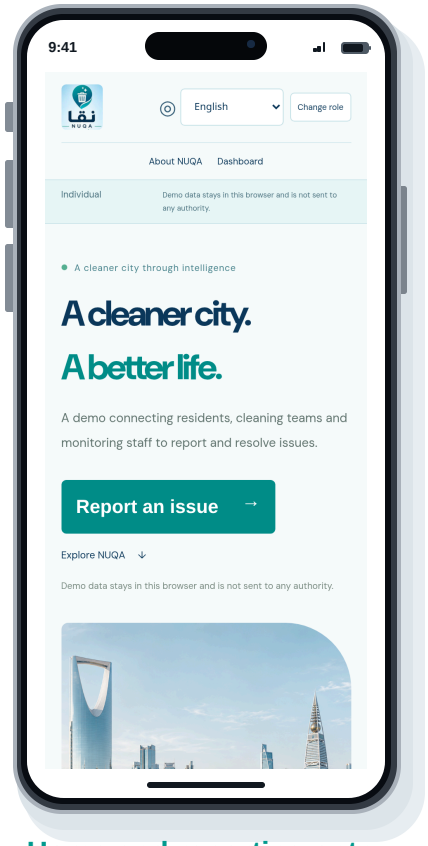
Start assigned work and submit completion for review.

03 Monitoring officer

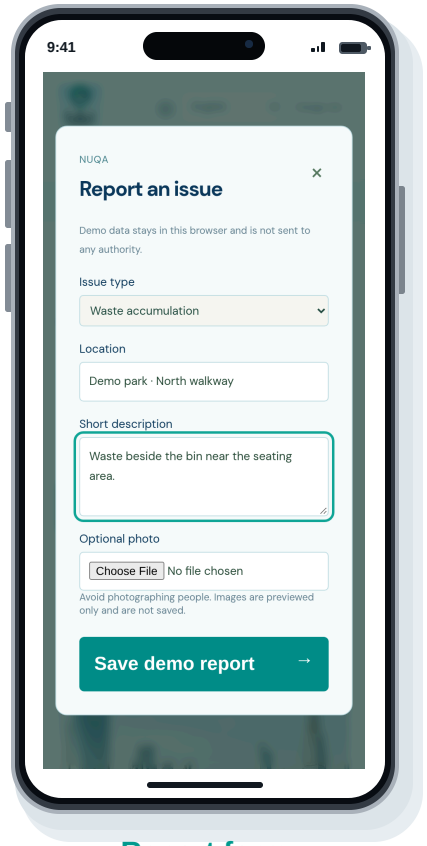
Review camera events, manage cases and verify closure.

Citizen reporting: simple inputs, visible progress

A short report connects a local observation to the same operational case record.



Home and reporting entry



Report form

01 Describe the issue

Choose the issue type, add a location and write a short description.

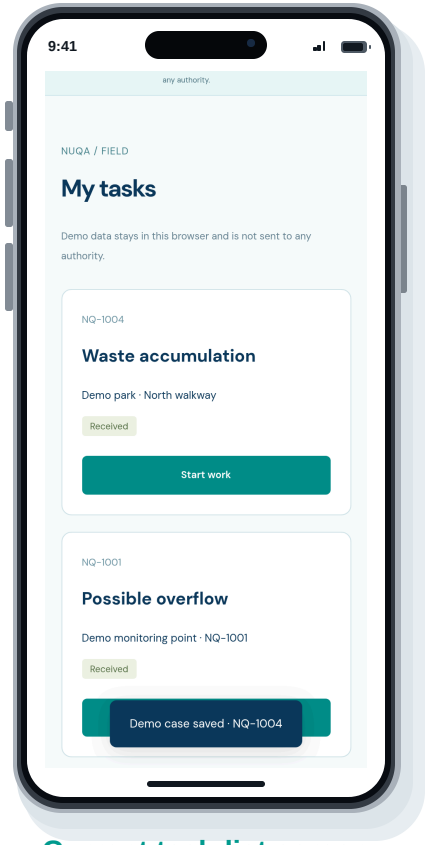
02 Preview, then submit

An optional photo adds context. Status updates keep the report trackable.

Current demo: reports stay in the browser; photos are previewed, not stored.

Sanitation teams: turn cases into accountable work

Tasks move forward with a defined completion handoff to monitoring staff.



Current task-list screen

01 Receive and start

A worker sees a task, issue type, location and current status.

02 Confirm completion

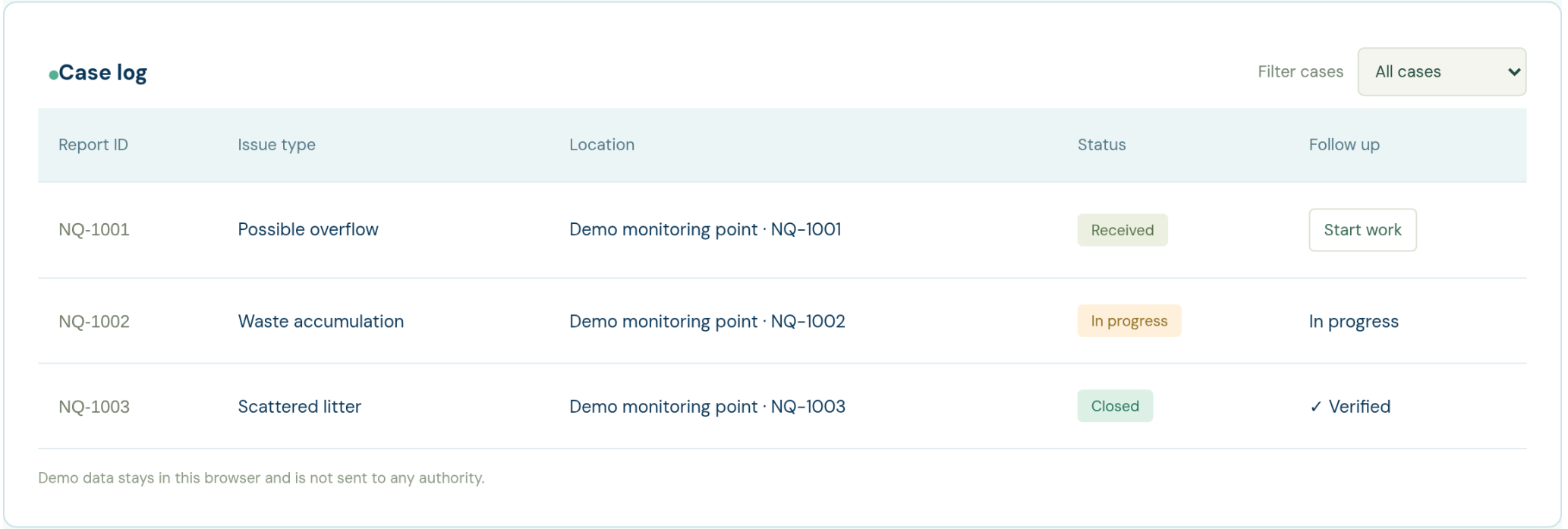
After cleanup, the worker submits the task for staff review.

03 Verify before closure

The officer checks the outcome and closes the case.

Municipal view: one record from report to closure

Staff can inspect status, initiate work and verify completed tasks.



Case log

Filter cases: All cases

Report ID	Issue type	Location	Status	Follow up
NQ-1001	Possible overflow	Demo monitoring point · NQ-1001	Received	Start work
NQ-1002	Waste accumulation	Demo monitoring point · NQ-1002	In progress	In progress
NQ-1003	Scattered litter	Demo monitoring point · NQ-1003	Closed	✓ Verified

Demo data stays in this browser and is not sent to any authority.

Unified cases

Camera events and citizen reports

Reviewed transitions

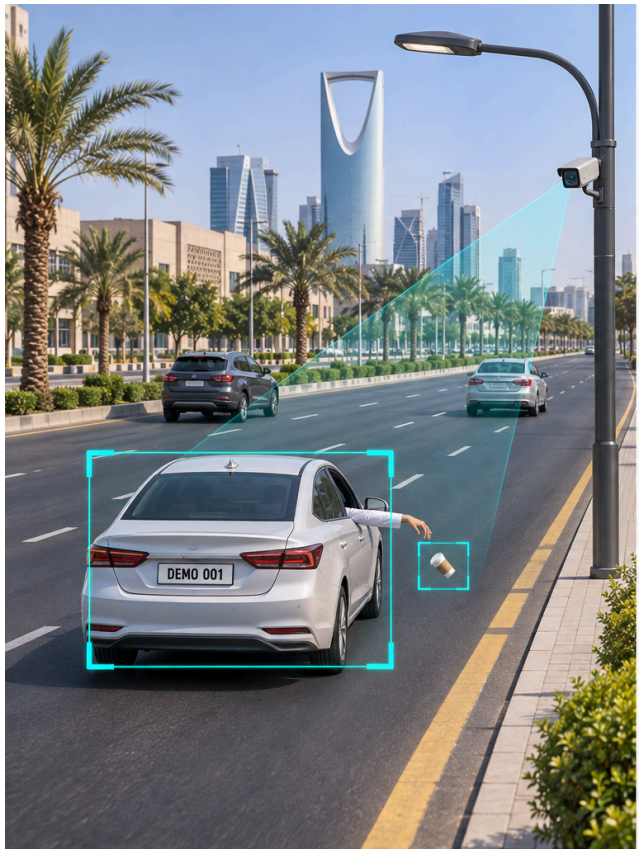
Received → In progress → Review

Verified closure

Officer confirmation before close

Public roads: detect a potential littering incident

AI analysis links a potential throwing event to a vehicle and a permitted plate snapshot.



Illustrative scene • Fictional plate • Proposed capability

01 AI detects the event

Track an object that may have been thrown from a vehicle.

02 Associate and read

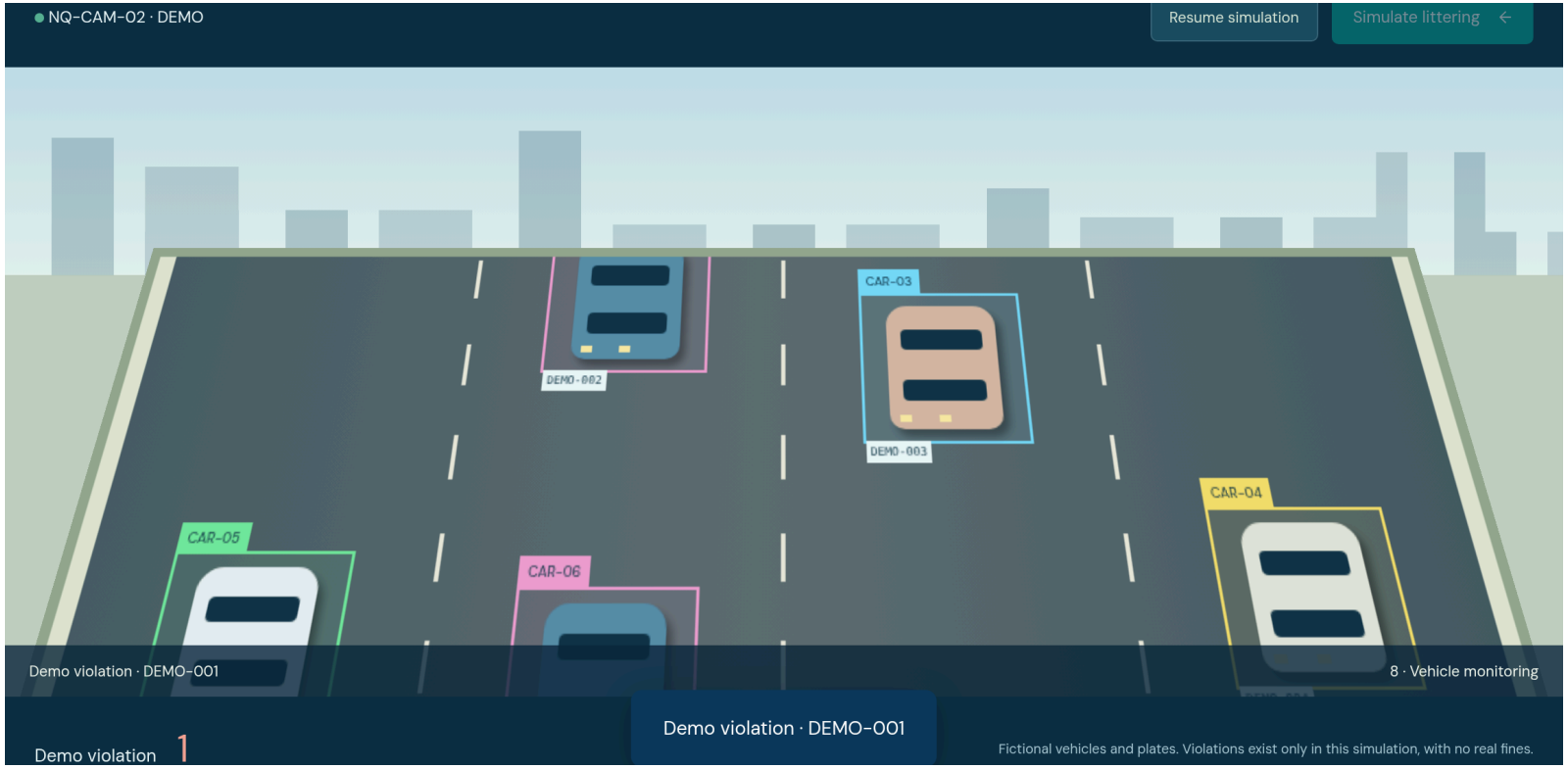
Link the event to the vehicle and use plate OCR when the image is readable and capture is permitted.

03 Submit for review

Attach location, time and selected evidence to a candidate incident.

Vehicle simulation: event → vehicle → demo record

The website illustrates the association using animated cars and fictional plates.



DEMO EVIDENCE

DEMO-001

Fictional vehicle plate

Demo violation record

Local record only.
No real fine is issued.

Implemented: a scripted event and local record. Proposed: real detection and authorized evidence review.

Parks and public spaces: cameras on lighting poles

Monitor waste hotspots around bins, paths and seating areas in an authorized pilot zone.

Use existing infrastructure

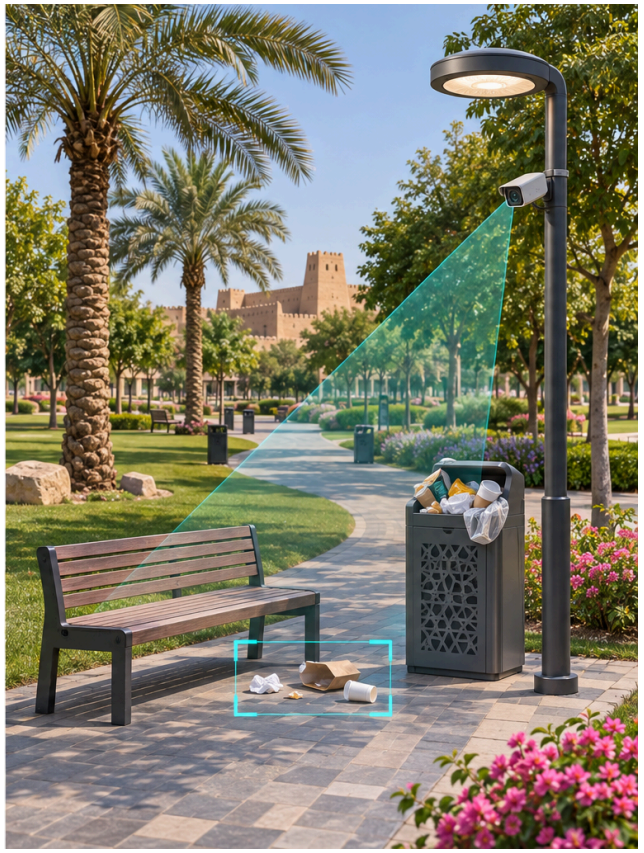
A proposed camera mounts below the lamp on an approved lighting pole.

AI analyzes the zone

Detect visible overflow, accumulated waste or a potential littering event near bins and benches.

Send a targeted response

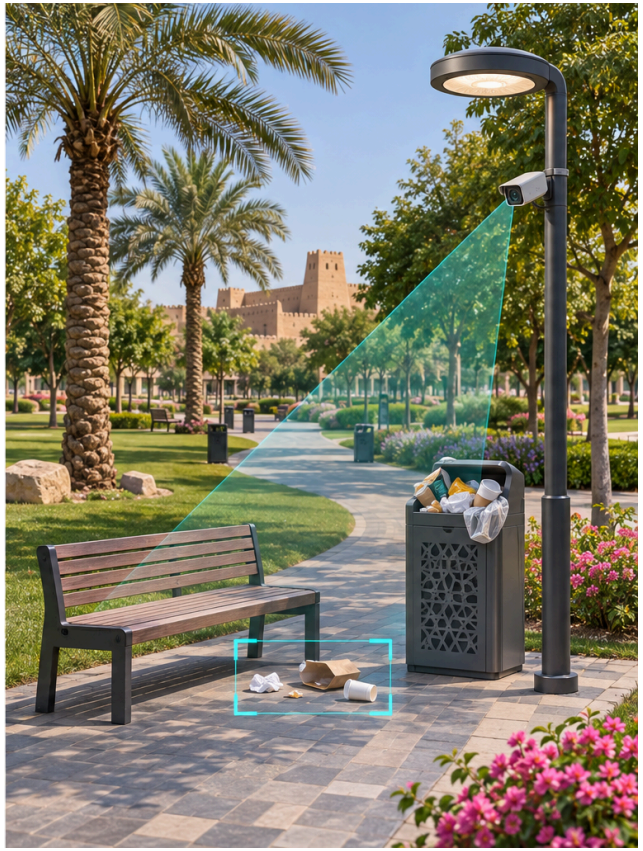
Create a reviewed task with a clear location for the cleaning team.



Illustrative park and camera coverage

Park simulation: AI flags litter near a seating area

A visual walkthrough of the proposed analysis, event record and cleaning response.



Illustrative scene and simulated event

PARK-DEMO-01 / LP-CAM-01

AI EVENT SIMULATION



LITTER-01

Detected condition

Scattered litter beside a bench

Event context

Camera zone + time + selected image

AI flag → Staff review → Cleaning task

Designed for the places people share

Concept settings for an authorized pilot, using cameras on existing lighting poles.



Public parks

Bins, lawns and seating areas



Walkways

Paths and shared pedestrian zones



Public plazas

Gathering areas and planted edges

From camera alert to cleanup: a shared workflow

Different sources enter one process; responsibility is visible at each handoff.

01

Detect

Camera or citizen

→

02

Review

Validate the event

→

03

Assign

Worker and task

→

04

Resolve

Clean and submit

→

05

Verify

Officer closes case

Potential vehicle littering → Evidence review → Authorized decision

Responsible evidence and human decisions

Design requirements for a future field pilot and any authority integration.

01

Purposeful capture

Focus on the incident. Avoid facial identification; mask unrelated people when appropriate.

02

Protected evidence

Restrict access to plates and images. Define retention, permissions and an audit trail.

03

Reviewed action

Check attribution and evidence quality. Allow correction before any real action.

Pilot plan: validate before deployment

Start with a small, authorized area and compare the system with field review.

01

Prepare

Choose road and park test zones. Confirm permissions and define a baseline.

02

Test

Evaluate events, vehicle association, plate visibility and usability.

03

Decide

Review false alerts, missed events, response times and verified cleanup.

Pilot measures Detection quality • Attribution errors • Response time • Task completion

The team

NUQA — A cleaner city through intelligence.

Manar Al-Nuqaydan

Tamadher

The next step

An authorized, controlled pilot that connects observation to a verified response.